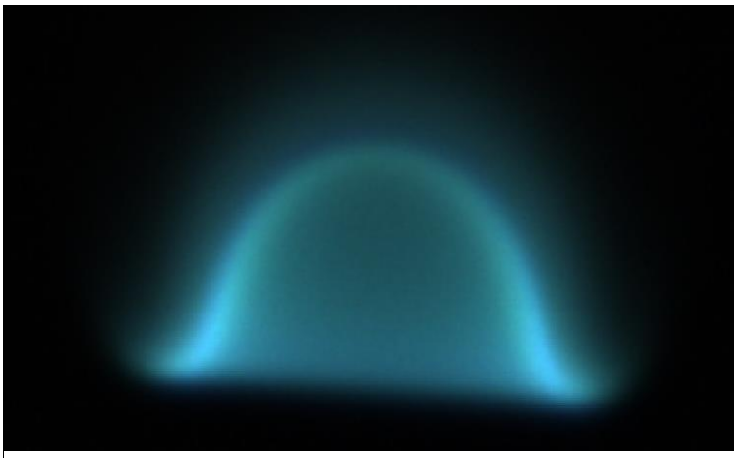
Advanced Combustion via Microgravity Experiments (ACME)

PURPOSE: ACME is focused on advanced combustion technology via fundamental microgravity research. The primary goal is to improve efficiency and reduce pollutant emission in practical terrestrial combustion. A secondary objective is fire prevention, especially for spacecraft.

RESEARCH: ACME includes six independent experiments investigating laminar, non-premixed flames of gaseous fuels. The experiments are being conducted with a single set of modular hardware in the Combustion Integrated Rack (CIR) on the International Space Station (ISS). While the ISS astronauts set up the experimental hardware, the ACME tests are remotely commanded from NASA's Glenn ISS Payload Operations Center.



Example image of a lifted coflow flame from the CLD Flame experiment led by Prof. Marshall B. Long (Yale University).

STATUS: ACME setup and checkout were carried out by the ISS crew in Sept.-Oct. 2017. Thus far, a dozen astronauts from Germany, Italy, Japan, Russia and the United States have supported the ACME research. Since November 2017, over 300 flames have been ignited for the Coflow Laminar Diffusion Flame (CLD Flame) and Electric-Field Effects on Laminar Diffusion Flames (E-FIELD Flames) experiments led respectively by Profs. Marshall B. Long (Yale University) and Derek Dunn-Rankin (University of California, Irvine).

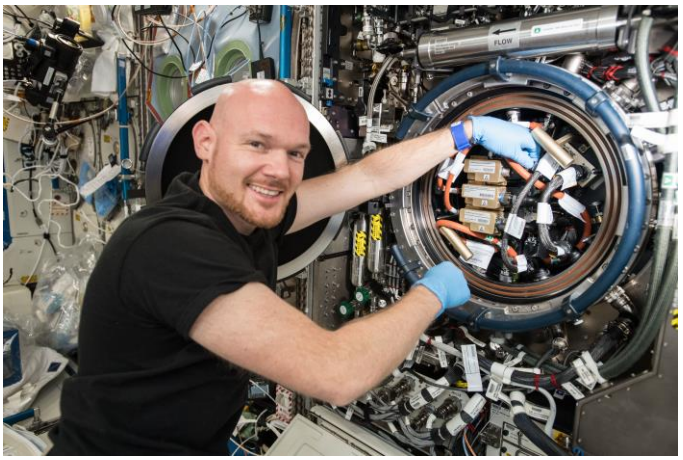
A second round of CLD Flame tests is currently underway and should end in September when the testing will transition back to the E-FIELD Flames experiment. In late 2018, the hardware will be reconfigured for the Burning Rate Emulator (BRE) experiment led by Prof. Jim Quintiere (University of Maryland). While the first two ACME experiments are focused on improving practical combustion on Earth, BRE is aimed at improving spacecraft fire safety.



Italian astronaut Paolo Nespoli installing the ACME chamber insert into the CIR's combustion chamber.



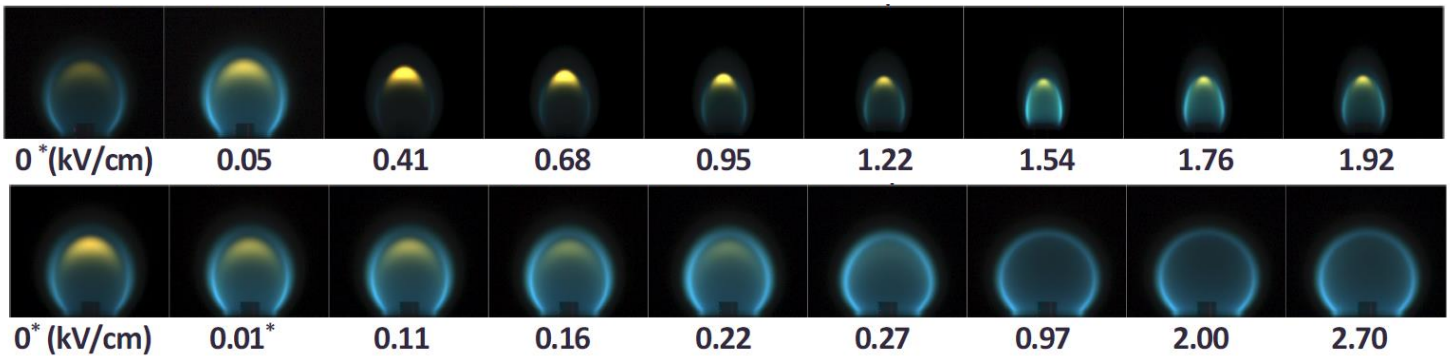
U.S. astronaut Joe Acaba aligning the retracting hot-wire igniter to the burner on the ACME chamber insert.



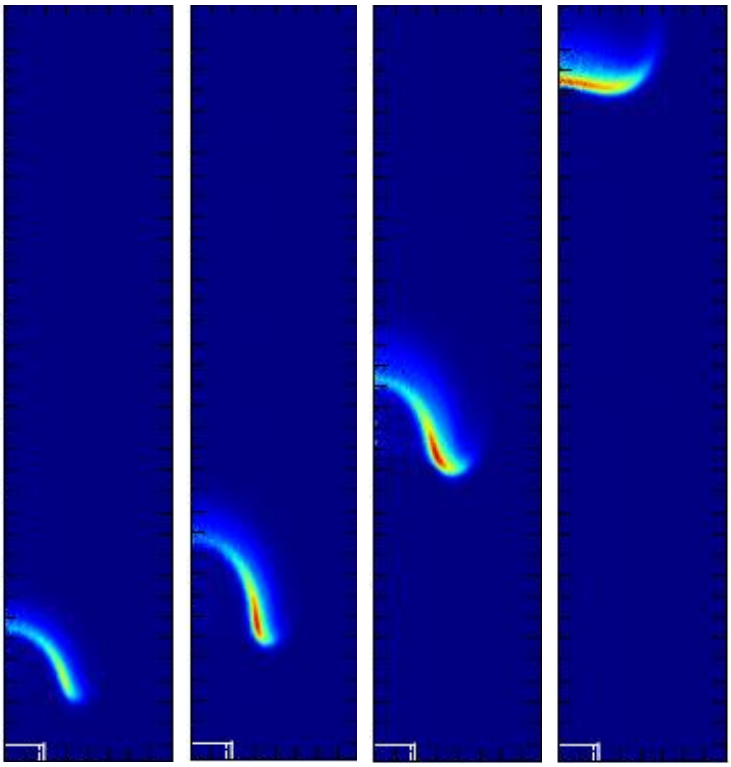
German astronaut Alex Gerst exchanging a mass flow controller (for the nitrogen used to dilute the gaseous fuel) on the ACME chamber insert.

20 August 2018

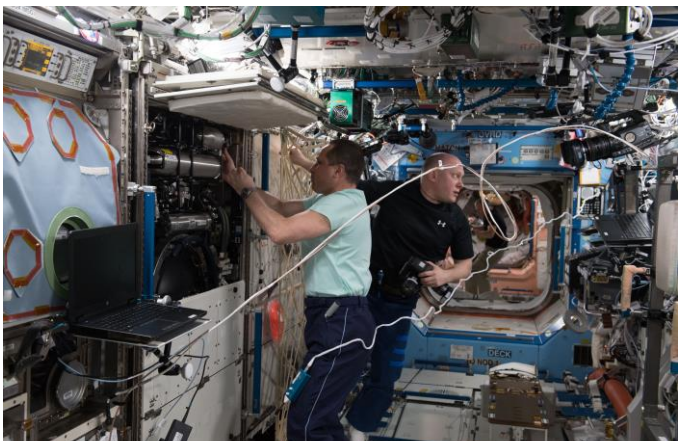
Advanced Combustion via Microgravity Experiments (ACME)



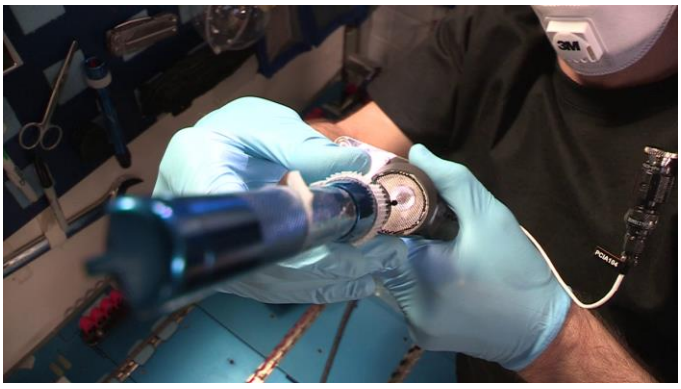
Diffusion flames with a weak air coflow as a function of the electric field strength, where an open copper mesh above the flame is negatively and positively charged in the top and bottom rows, respectively. Flames naturally include ions which can be affected by a powerful electric or magnetic field. In ACME's E-FIELD Flames experiment, the copper mesh is energized up to 10,000 volts. This technology could be used to stabilize fuel-lean flames under conditions which are less polluting.



Lifted flames (detached from the burner) as a function of the matched fuel and air velocity from the CLD Flame experiment. The false color indicates the CH* concentration as function of the radius and height above the burner. CH* is an excited species and emits blue light as it drops to a lower energy level in a process known as chemiluminescence. It is the primary source of the blue that is the natural color of hydrocarbon flames (neglecting the broadband thermal radiation from the soot).



Russian cosmonauts Anton Shkaplerov (left) and Oleg Artemyev exchanging a CIR gas bottle.



U.S. astronaut Ricky Arnold cleaning soot from the coflow burner used by both the CLD Flame and E-FIELD Flames experiments. The cleaning was carried out with a toothbrush, piece of insulated wire, vacuum sweeper, flashlight (for burner inspection), dust mask, and gloves.

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